LINEAR ALGEBRA · LESSON 14

Projections onto Subspaces

Can't hedge it exactly? Get as close as possible. The **best possible hedge** is a projection.

$A^TA\hat{x} = A^Tb$ hedge ratio = β MIT 18.06, Lecture 15

ONTO A LINE

Drop a perpendicular

The error $e = b - \hat{x}a$ must be perpendicular to a:

$$a^T(b - \hat{x}a) = 0 \;\Rightarrow\; \hat{x} = \frac{a^T b}{a^T a}$$

$p = \hat{x}a$ is the closest point on the line; $P = aa^T/a^T a$, with $P^2 = P$.

The hedge ratio is beta

Nifty $a = (-20, 0, 20)$, midcap $b = (-26, 2, 24)$:

$$\hat{x} = \frac{520 + 0 + 480}{400 + 0 + 400} = \frac{1000}{800} = 1.25$$

midcap $= 1.25\times$ Nifty $+(-1, 2, -1)$. Leftover $\perp$ Nifty: $20 + 0 - 20 = 0$.

MINIMUM RISK

Why it's the best

Leftover risk $\|b - ha\|^2$ is a parabola in h. Its bottom is exactly where the leftover is perpendicular to the hedge: $h = \beta = 1.25$.

> HEDGE
>
> Short ₹1.25 Nifty per ₹1 midcap. The leftover $\|e\|^2 = 6$ (vs 1256) can't be hedged with Nifty.

The normal equations

$$A^T(b - A\hat{x}) = 0 \;\Rightarrow\; A^T A\,\hat{x} = A^T b$$

$$p = A\hat{x} \qquad P = A(A^T A)^{-1} A^T$$

The error lands in the left null space: $b = p + e$, $p \in C(A)$, $e \in N(A^T)$.

THE CLOSEST TARGET

(10, 105, 4) is impossible

Nifty + bonds: $A^TA = \begin{bmatrix} 146 & 85 \\ 85 & 50 \end{bmatrix}$, $A^Tb = (1274, 745)$.

$\hat{x} = (5, 6.4) \rightarrow$ closest reachable target $(11.4, 104.8, 5)$.

Error $(-1.4, 0.2, -1) = -0.2 \times (7, -1, 5)$: along the hidden link.

HEDGING

Unhedgeable risk = the error

Exposure $(10, 3, 2)$ with only Nifty and IT futures: $\hat{x} = (7, 3)$, hedged part $(10, 3, 0)$, leftover $(0, 0, 2)$: the rate risk nothing can touch.

Payoff $(-100, 50, 300)$ with futures + call: $(0.5, 1)$, leftover $(0, 50, 0)$.

RECAP

Key takeaways

1. Line: $\hat{x} = a^T b / a^T a$, $P = a a^T / a^T a$.
2. Subspace: $A^T A \hat{x} = A^T b$.
3. $P^T = P$, $P^2 = P$.
4. Error $\perp$ subspace, in $N(A^T)$.
5. Best hedge = projection; ratio = β.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A hedge-ratio slider with the risk curve, a projection lab, and worked hedging and options examples:

learn.krisarj.com

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